The Militarization and Weaponization of Outer Space and How It Will Affect International Relations

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"*What we have to be careful about as a nation is the weaponization of space. Space is critical to our country, to our commerce, and day-to-day life."*-General Mark Milley, Chairman of the Joint Chiefs of Staff

The United States of America, the Russian Federation, and the People's Republic of China (PRC) are considered to be the hegemons of outer space with the largest military and commercial presence in Earth's orbit. These nations are also overpassing the international treaties and norms that have governed outer space for the past sixty years. Other space-faring states, like the Republic of India, Japan, the European Space Agency, etc., will not be considered dominant in outer space due to their level of funding, number of satellites, and lack of ambition in space policy on militarization and weaponization. The United States of America, the Russian Federation, and the PRC are the dominant players in outer space, granting them the power and authority to control space policy and operations. The militarization and weaponization of outer space have invalidated international space treaties, which will lead to a rise in relational tensions, competition, and a space arms-race between space-faring nations.

Treaties Governing Outer Space

In 1967, the Outer Space Treaty was ratified by the General Assembly of the United Nations, marking the first international treaty regarding outer space in human history. The Outer Space Treaty is "a diplomatic agreement that outlawed nuclear weapons from space, forbade military installations and manoeuvres [sic] from the Moon and other celestial bodies, and declared the cosmos a special zone free for all to use and explore," (Bueno, 2020). Both the United States of America and the former Union of Soviet Socialist Republics (USSR) advanced the treaty to be installed in the United Nations. The Russian Federation is the successor state to the USSR and replaced the USSR in the United Nations, making Russia the DeFacto signatory in the treaty. The United Nations General Assembly later passed the following shortened titled resolutions: the ‘Rescue Agreement’, the ‘Liability Convention’, the ‘Registration Convention’, and the ‘Moon Agreement.’ The treaties established by the United Nations considered smaller, non-space-faring nations that did not have the capability, nor funding, to launch a satellite into outer space, let alone create a space agency.

The Outer Space Treaty of 1967 established the foundation of space law. Bueno (2020) states that the agreements in outer space took much inspiration from the established treaties of the high seas; a zone free for use and exploration for all humans. He noted the significance of the wording in ‘the province of all mankind’ (Bueno, 2020), meaning that every human being on Earth shares the right to outer space; No state shall claim outer space. According to the United Nations Office of Outer Space Agreements (UNOOSA, n.d.), the Outer Space Treaty comprises of 17 Articles that flowed in overtime with additions of new treaties to amend the existing Outer Space Treaty. On 3 December 1968, the Rescue Agreement requires all States take steps to rescue and assist astronauts in destress and aid in recovering space objects that return to Earth. On 1 September 1972, the Liability Convention entered into force, which provided a state should be held absolutely liable to pay compensation for damage caused by objects that it placed in space falling down to Earth or another state’s object in space. (UNOOSA, n.d.); Therefore, space debris caused by a State that damages a satellite of another State will be held liable for damages caused by said debris under international law. With the tracking of space debris, a state who caused damage due to space debris will be held liable, due to its negligence. On 15 September 1976, the Registration Convention entered into force which provided all “States with a means to assist in the identification of space objects" (UNOOSA, n.d.) to track as many space objects as possible. All space-faring nations have space tracking centers, but the United States Space Surveillance Network is the largest network providing alerts and data regarding space debris to all states that may need it. The Moon Agreement entered into force on 11 July 1984, providing all celestial bodies "be used exclusively for peaceful purposes, that their environments should not be disrupted, that the United Nations should be informed of the location and purpose of any station establish on those bodies." (UNOOSA, n.d.); In addition, the natural resources of the Moon are to be shared amongst all humankind, therefore, an 'international regime' should be established to govern the exploitation of such resources when feasible. This specific agreement will become relevant further on due to the specific mention of the exploitation of resources.

There are five declarations and legal principles that the United Nations has implemented regarding outer space. According to UNOOSA (n.d.), the Declaration of Legal Principles Governing the Activities of States in the Exploration and Uses of Outer Space created the skeleton for the Outer Space Treaty, setting up the basic legal framework for the treaty governing outer space. The rest are as follows: The principles Governing the Use by States of Artificial Earth Satellites for International Direct Television Broadcasting; The Principles Relating to Remote Sensing of the Earth from outer Space; The Principles Relevant to the Use of Nuclear Power Sources in Outer Space; and the Declaration of International Cooperation in the Exploration and Use of Outer Space for the Benefit and in the Interest of All States, and Taking into Particular Account the Needs of Developing Countries. The 'Broadcasting Principles' and the 'Remote Sensing Principles' are in place to set up rules about how to use communication, television, and global positioning systems (GPS) satellites. The 'Nuclear Power Sources' principles goes into great detail from specifically what nuclear reactors/power sources are allowed in space to what radioactive fuel is allowed to be used. The 'Benefits Declaration' reinforces the notion that space should be used for the benefit and advancement of all humankind (UNOOSA, n.d.).

These agreements and principles are all subject to international law, where countries come together to follow an international laws-based order to enforce international norms. Although the United Nations has implemented these resolutions and laws, multinational organizations do not have the authority to enforce them upon individual nations. These space treaties and principles are, in turn, not legally binding. These treaties have set up rules on how states shall operate in outer space, especially regarding the weaponization of outer space; but nations are less incentivized to follow these rules anymore. When the Outer Space Treaty was created, the prospect of weaponizing space was impractical because the technology was not there. Although, "[h]ad the technology been available to nuclearise [sic] and weaponise [sic] space effectively, many scholars agree the OST would never have materialised [sic]." (Bueno, 2020). As can be seen today, space-faring states are testing the limits or completely disregarding the limitations set up by the Outer Space Treaty. The militarization of outer space has already occurred, however, due to recent technological advances, the weaponization of space is coming to fruition. Technology is advancing too fast for laws to be negotiated formulated to govern these new capabilities.

When creating the Outer Space Treaty, the USSR and the United States wanted to have an understanding that the other would not try to exploit, claim, or weaponize space (i.e., placing nuclear weapons in space) to prevent unnecessary competition and an expensive arms race. They came to this agreement because the technology was not there to exploit, claim, or weaponize. Now, the technology is here to exploit and weaponize space. When one nation begins to weaponize space that jeopardizes the national security of another nation, the other nation will invest in the militarization and weaponizing of space. Treaties that are not legally binding are lower on the list compared to their respective national securities.

Multinational Militarization and Weaponization of Outer Space

"*Slowly but surely, we are heading toward [militarization of space]. Roscosmos has no illusions about this. Everyone is working on it*." -Dimtry Rogozin, Chief Russian State Corporation Roscosmos

The militarization and weaponization of space was inevitable; just as how the militarization and weaponization of land, sea, air, and cyberspace was inevitable. The militarization of a domain does not necessarily include weapons, it more aligns to the supporting of military operations and use of that domain by a military body; Equipment like the GPS or the similar Russian GLONASS, which helps track and guide missiles, or cameras in space, which can locate adversary locations. Navigation satellites can guide civilian systems as well as military systems/platforms with "precision accuracy and serve as a force multiplier for ground military operations" (Pražák, 2021).

Space has been militarized ever since Sputnik was launched in 1957, where the satellite used radio signals to send back to the USSR. Sputnik was not a weapon, but the rocket used to escort Sputnik to orbit could easily escort a nuclear warhead across the planet. Weaponization deals directly with weapons, kinetic and non-kinetic. One can militarize an aircraft by using it to spy on an adversary, but the aircraft becomes weaponized once a bomb is strapped on to drop it on an adversary. Weapons in space range from the use of electronic warfare (radio jamming, hacking, etc.) to laser and plasma weapons. It sounds like something out of a science fiction piece, yet these weapons have been implemented by militaries and are in practice. The nations that have invested the most in developing military capabilities in outer space are the United States of America, the Russian Federation, and the PRC. Noting India, who has conducted a successful anti-satellite (ASAT) missile test, but they will not be evaluated. For these states, "mastering space technology has become an essential factor of power and sovereignty. It has the same importance today as that of deterrence in the 1960s" (Montluc, 2011).

The United States of America

The United States of America, currently, has the most militarized presence in outer space; therfore, they have the most to lose. They have an "integrated military approach to space to increase warfighting capabilities" (Grossman, 2000), resulting in most of their military being heavily reliant on space. This is one of the justifications to increase their budget in space operations; to maintain conventional readiness and obtain an advantage in this new domain. If an adversary of the United States were to eliminate their satellite network or disrupt their communication relays, then the United States military would drastically lose its notorious lethality and effectiveness. This has the United States to prioritize funding towards platforms like the U.S. Aegis Ballistic Missile Defense System, which can utilize ASAT capabilities. According to Pražák (2021) Presidents like Ronald Reagan invested in space-based anti-ballistic missiles interceptors to protect against nuclear ballistic missiles. Notoriously, the Reagan Star Wars Program would see the rapid advancement of laser technology (Pražák, 2021), but not implemented in the way that the aging president desired it to be. The official militarization of outer space from America came when former President, Donald Trump, announced the formation of the newest branch of the United States Military: The United States Space Force.

The Russian Federation

The Russia Federation, and its predecessor, the USSR, originally was the most aggressive in developing outer space capabilities. After the splintering of the Soviet Union, the Russian Federation came out as a shell of its former power and ability to keep up technologically to the United States and China in outer space. They could no longer afford to allocate resources to programs like its fractional orbital bombardment system, which was “a weapon designed to orbit a nuclear warhead and de-orbit it over a target of choice.” (Bueno, 2020). The Soviet military successfully tested this weapon, but never advanced beyond tests, due to nuclear weapons being banned in space. Yet, it did not deter them from testing.

“On November 15, 2021, the Russian Federation recklessly conducted a destructive test of a direct-ascent anti-satellite missile against one of its own satellites.” (United States Department of State, 2021), which caused an extraordinary amount of space debris to be created that still loiters in low-earth orbit. This showed strength, to show that it can eliminate adversarial satellites. According to Jackson (2020), Russia’s outer space policy is to assert Russia’s authority, status, and prestige while reviving its economy. The economic, military and technological weakness of Russia has led them to pursue asymmetric tactics such as working within global institutions, such as the U.N., to gain legitimacy in outer space (Jackson, 2020). Although, due to the war in Ukraine, Russia’s legitimacy on the global stage has diminished to a pariah state, showing its military failures and incompetencies. They still have significant capabilities; it is their reputation that can no longer hold up.

The People’s Republic of China

The PRC is the fastest growing space power out of the three contending powers. According to Gubrud (2011) the PRC conducted its first ever direct ascent ballistic ASAT missile test in 2007, catching the attention of the United States (Gubrud, 2011). The PRC military has been very involved in outer space, saying this:

> Space... a critical factor for deciding military transformation; and it has an extremely important influence on the evolution of future form-states, modes, and rules of war. Therefore, following with interest the military struggle circumstance of space and strengthening the study of the space military struggle problem is a very important topic we are currently facing. (Defense Intelligence Agency, 2022, pg. 8).

Chinese space policy dictates that it will militarily dedicate more towards the space domain. It is ready to invest in the militarization and weaponization of space. President Xi Jinping stated, "To explore the vast cosmos, develop the space industry and build China into a space power is our eternal dream." (The State Council Information Office of the People's Republic of China, 2022), space industry is vital to their overall national strategy.

China has invested heavily into their outer space capabilities, be it militarization or weaponization. Their most iconic innovation is their active debris removal (ADR) system, which is a satellite with robotic arms, claimed to be "one of the most effective way[s] to deal with... uncontrollable space objects" (Xu, Yan, Hu, Liang, 2019). According to Xu, et al. (2019) This new system has sophisticated tracking and capture mechanisms that increases the effectiveness to deal with these uncontrolled space objects by capturing them and de-orbiting them (Xu, et al., 2019). Even though this system was created for the purpose of removing dangerous debris from the orbit of Earth it is an example of a potential space weapon with the capabilities of de-orbiting functional systems, such as lasers, GPS, or communication satellites. This system is the precursor to future interception systems which will ultimately lead to the development of the science of space warfighting and potential space dogfights.

Due to the intense militarization and weaponization of space by the U.S., Russia, and the PRC, space has become an integral part of their respective militaries. If a war were to break out between any of these states, it would first take place in the space domain. It is absolutely vital for each state's survival to heavily invest in outer space capabilites. These new capabilities, theoretically, will have the same deterrence effect as nuclear weapons had in the 1960's and today. Though, to have this deterrence, an arms race would occur.

The Space Arms-Race

"We didn't go to the moon to explore or because it was in our DNA or because we're Americans. We went because we were at war and we felt a threat" -Neil deGrasse Tyson

"An arms race occurs when two or more countries increase the size and quality of military resources to gain military and political superiority over one another" (History.com Editors, 2009). An arms race is caused by fear of a conflict with the other power due to a new, advanced weapons platform, such as, the dreadnought arms race between the German Empire and the British Empire in the early 20th century, the Cold War arms-race for atomic bombs, and now, a blooming arms-race in space between the United States, Russia, and the PRC. Unlike previous arms races, like the UK-German dreadnought arms race, or US-Soviet nuclear arms race, this will be a "technologically driven arms race" (Sönnichsen & Lambach, 2020, pg. 1). All of these arms races have led to all led to war; the proliferation of space will lead to conflict, as well. These three states will invest in relatively conventional style weapons compared to the most recent nuclear arms race. These new weapons will define this new and developing age of warfare. More importantly, there will be a Cold-War style ‘space-race’, where these great powers will try to show off whose stick is bigger by accomplishing technological and scientific missions on celestial bodies, such as the Moon and Mars.

Advanced technology weapons will be used in the future, and it will require substantial resources and capital to invest in such platforms. According to Sheposh (2023), there is development of ASAT weapons, kinetic-kill vehicles, high-energy chemical lasers, enhanced cybernetic capabilities, and particle beams. Chemical lasers produce energy when they interact with atoms in a chemical substance that then can be targeted toward satellites to neutralize them. Particle beams are weapons that fire subatomic particles at the speed of light to destroy satellites (Sheposh, 2023). There are no treaties or agreements about these weapons. These three states will not choose to trust the other’s ‘good-will’ in not developing these weapons, because there is no incentive not to; the only incentive is to develop more weapons systems in the name of national security. The proliferation of outer space with information systems will, according to Arbatov and Dvorkin (2010), develop into light, rapidly deployable spacecraft. This will be increasing the survivability of such platforms, thus operability (Arbatov & Dvorkin, 2010)

The State Council Information Office of the PRC (2022) has stated that the PRC will be heavily investing in outer space capabilities. They plan to establish a scientific lunar base within the next ten years and have humans touch-down a few years after. There is production of the rocket that will transport Chinese assets to the Martian surface, as well (The State Counsil Information Office of the PRC, 2022).

The United States has been increasing the amount of military spending on outer space capabilities. The National Aeronautics and Space Administration (NASA, n.a.) founded Artemis, which is the name of the program that was created to establish a lunar base and re-visit the moon with organic bodies (humans). The Artemis accords are in collaboration with a twenty-two U.S. allies (NASA, n.a.). According to Arbatov and Dvorkin (2010), the U.S. rejected a United Nations proposal to ban all weapons in outer space. They justified this by stating they reject legal regimes seeking to prohibit or limit access to or use of outer space. They would rather advocate preventing an arms race in space by continuing dialogue and enhancing transparency to increase trust. (Arbatov & Dvorkin, 2010).

These three superpowers are currently attempting to surpass each other's capabilities for their own respective strategic and political gains. Russia is notorious for its military posturing and nuclear saber rattling. The President of the Russian Federation, Vladimir Putin, has stated the following:

> In the new, 21st century, Russia must maintain its status as a leading nuclear and space power because the space industry is directly linked with defence [sp] and I would like to remind you of this. Today, we will discuss issues related to long-term priorities of space exploration and will analyse [sp] what we must do to strengthen our positions in this truly strategic area. (Defense Intelligence Agency, 2022, Pg. 20).

Russia will try to keep pace with the United States and the PRC in space technology and capabilities.

Effects on International Relations

Any action that a State carries out will garner a positive, negative, or neutral reaction from all sovereign States across the world. A lack of diplomatic and political finesse will tarnish international relations and attitude towards one’s State. The foundation of the United States Space Force and “Trump’s insistence that the United States achieve ‘dominance’ in space... has unnerved people all over the world, Chinses and Russian officials especially.” (Bueno, 2020).

The space domain has created new challenges for these three space faring superpowers. Growing hostilities and suspicion towards one another are causing relations to sour. Although there are efforts of multilateralist cooperation, the underlying fact is that each nation's actions in space have specific effects on their respective international relations.

PRC-U.S. Relations

The United States and the PRC are political and military rivals. China's growing military threat to East Asia and Sount East Asia has caused the United States to bolster its posture in that theater, even in the wake of the Ukraine-Russo War. In 2011, the United States Congress passed the Wolf Amendment, which severed scientific ties between NASA and the PRC. This was passed due to the concern that China would steal U.S. space technology, which is a valid concern given China's theft of other technologies. The Wolf Amendment essentially severed a source of multilateral communication. Whitford (2019) points out that this amendment would increase the chance of a war in space, sending a signal to China that the U.S. does not see space as a multilateral environment with the intent of domination. With the ever-rising importance of satellites in the modern era and modern militaries, a space conflict could arise when one side assumes the other to threaten their satellites; therefore, threatening national security. This is why there is a hotline between Beijing and Washington with the intention of preventing a space war (Whitford, 2019). The U.S. and the PRC know that their relations are extremely strained and growing in hostility, yet they understand how devastating it would be to lose essential space assets that contribute to everyday life. The connection of a hotline between the two space superpowers is reminiscent of the hotline between Washington and Moscow during the Cold War.

Economic ties with China seem to be the most reliable support to maintain relations, even with the outset of the U.S.-China Trade War in 2017. Since these two are the most powerful nations on Earth and space, “Henry Kissinger notes, ‘[China and the United States] would do well to recognize that their rhetoric, as much as their actual policies, can feed into the other’s suspicions.” (Burris, 2013, pg. 110). As previously stated, it would be devastating to lose satellites that contribute to everyday life in the modern age; the same can be said with risking economic ties between the United States and China. These nations have very interdependent economic relations that would collapse each other’s economies if war broke out. The two nations should take note of the effects their actions might have on their relations. Any ambitious change in the space domain from China will cause concern for the United States. Any aggressive change in the space domain from the United States will cause a similar change in China. This will cause a feedback loop, leading to increasingly aggressive policy changes. The glue that keeps relations intact is their economic ties.

PRC-Russia Relations

Historically, they have been regional rivals, even ideological rivals during the Cold War. Yet, now, relations between China and Russia are mutually beneficial and situational. Their close ties are due to their competition with a common enemy: The United States, the PRC and Russia have very similar rocket designs, as can be seen with their radial booster design with the iconic slope towards the center of the rocket. It is not the first time that China has copied Russian designs and massively improved upon them. Regarding the Moon, “China launched the international lunar research station project together with Russia and initiated the Sino-Russian Joint Data Center for Lunar and Deep-space Exploration.” (The State Council Information Office of the People’s Republic of China, 2022). In sum, their relations are friendly cordial, due to their competition with the United States.

Both, China and Russia, see the United States as an enemy. As many politicians, generals, and philosophers have noted, ‘the enemy of my enemy is my friend’. According to the Defense Intelligence Agency (2022), China and Russia seek to exploit the perceived U.S. reliance on space-based systems and challenge its position in the space domain. China and Russia are intending to become the leading space powers and create new global space norms. They aspire to undercut U.S. leadership through space and counterspace capabilities. (Defense Intelligence Agency, 2022). In sum, cooperation between China and Russia is mutually beneficial because it allows them to hamper U.S. gains in outer space while working together in advancing their space technology. The U.S. would have to increase their energy and spending on space, while China and Russia can rely on one another in space and focus energy and resources somewhere else.

U.S.-Russia Relations

For the past sixty years, the U.S. and Russia have negative and positive international relations. Russia’s predecessor, the USSR, and the United States were rivals during the Cold War, yet they came to many multilateral agreements that prevented hostilities, like the Outer Space Treaty and the START/New START Treaties. [INSERT SOURCE] The nuclear nonproliferation treaties were created due to the two nation’s contesting nuclear interests, specifically in nuclear weapons. One of the first treaties regarding the use of nuclear weapons was the Outer Space Treaty, which bans weapons of mass destruction from being placed in space. Treaties and agreements take much effort and years of building trust, yet it takes very little to dismantle said treaties and agreements. When a new domain could prove to grant an unfair military advantage to one nation over another, tensions rise in tandem with the technological exploitation of this domain. Space is the new domain, and relations are, again, poor between Russia and the United States.

The only time in the past century where Russia and the U.S. were friendly was the 20-year period after the USSR fell. Russia has recently made multiple provocative actions towards the U.S., Pražák (2021) notes the January 2020 incident where two unknown satellites from Russia, likely with inspection capabilities, intentionally orbited closely to an American reconnaissance satellite (Pražák, 2021). The presumption is that these satellites are "'assassin' craft designed to get close to US satellites and destroy them" (Sheposh, 2023). This more aggressive posture in space from Russia has, according to Sheposh (2023), caused U.S. policy to be pushed toward more offensive capabilities. Now, the U.S. will take a more active role in defending its interests in outer space (Sheposh, 2023). Good-faith treaties and norm setting cannot be created when these superpowers continue to increase their posture in space.

US policy has also begun to shift toward more offensive capabilities and taking a more active role in defending its interests in outer space. According to Montluc (2011), dual-use technologies, such as earth observation technology can be used to map floods, monitor deserts, and look at the state of oceans or polar icecaps. Earth observation technology can also be used to observe for defense and security purposes, being the eye in the sky to accurately target enemy positions (2011). Relations are at a Cold-War low between Russia and the U.S., especially with U.S. weapons and space assets assisting Ukraine in their war against Russia. This is unique, for Russia cannot destroy American satellites without retaliation, although America can share with Ukraine their intelligence, surveillance, and reconnaissance (ISR) capabilities to give them an upper hand on the battlefield against Russian forces. This places Russia in a lose-lose position and the U.S. vice-versa.

Future Flashpoints for Conflicts

"I don't want to go into space because of war. I think we would if it were triggered. If China said they want to put military bases on Mars, we'd be at Mars in two years. That would be quick. I don't want that to be the reason." -Neil deGrasse Tyson

The thought of a war with Russia or the PRC is worrying, although, "war with China is not inevitable whether for control of outer space or otherwise." (Burris, 2013, pg. 108). The most likely scenarios for a war to begin are celestial body territorial dispute over resources (such as water), a space debris accident, or a significant non-kinetic attack.

If water is discovered on Mars or the Moon, there will be competition to see who can arrive there first. If it is contested, and diplomatic avenues collapse, a space war could spark. Water will be the most valuable resource in space colonization and space travel. Water is comprised of two hydrogen atoms that attach to a single oxygen atom. When these atoms are separated, they make regular hydrogen and oxygen. Oxygen supports life, and hydrogen can be turned into fuel for engines on spacecraft. Once these states have the technology and capability, they will establish outposts and bases to secure this highly valuable resource. As a matter of fact, the Defense Intelligence Agency (2022) notes the Russian Luna-27 mission to establish a base on the moon over a frozen water deposit by 2025. China plans to become the second nation to send humans to the moon by 2030 and establish a research base by 2050 (Defense Intelligence Agency, 2022). The United States plans to "extract and utilize resources on the Moon, Mars, and asteroids" (NASA, n.d.). If these nations do not follow the Outer Space Treaty's articles relating to resource extraction, they must find a way to cooperate and designate a shared resource policy.

Since space debris is so well documented and tracked, an adversary's debris destroying a multi-million-dollar satellite is devastating and might lead to retaliation. Pražák (2021) states that the majority of space debris resides in low-earth orbit, which is the most occupied orbit with satellites; the high speeds of the debris pose a real danger to other systems. The most dangerous proliferation of orbital debris in outer space are ASAT weapons (Pražák, 2021) that will destroy satellites and caused untold amounts of debris that will create a ripple effect of debris causing more debris.

A significant non-kinetic attack, such as electronic warfare use or a cyber-attack, that is damaging enough could start a conflict. A significant, damaging scenario would look like a state targeting another state's power grids that connect to critical infrastructure (i.e., water, heating, electrical systems, communication, etc.), which would cause millions of dollars of damage, and would lead to deaths. Hospitals would run out of essential power to keep critical patients alive. During winter, heating keeps millions warm; turning off the heat would lead to injury and death. To reinforce, "cyber operations or cyberattacks may cause more severe damage." (Pražák, 2021). Cyberattacks have taken place between all three powers, but none as large as the scenario presented. There is no convention on the rules of cyberattacks.

Conclusion

A space war is not inevitable, but the laws governing space are no longer being taken as seriously as they used to. The Outer Space Treaty designated outer space to be a place for peaceful cooperation for the benefit of all humankind, yet, Russia, PRC, and the United States are selectively deciding which rules to abide by and which to flaunt. The militarization and weaponization of space have invalidated these treaties regarding outer space. Satellites are a pivotal arm of terrestrial militaries, ushering in space as the latest military domain. There is an active arms race in space to see which nation can proclaim preeminence in outer space. The rapid development in outer space weapons, such as electronic warfare capabilities, ASAT missiles, chemical lasers, and plasma weapons will create a chain reaction of proliferation. Arms races increase global political tensions and limit diplomatic cooperation. Relations may change due to terrestrial politics, but these three states will remain rivals for the foreseeable future.

References

Arbatov, A & Dvorkin, V. (2010). *Outer Space : Weapons, Diplomacy, and Security*. Carnegie Endowment for Int'l Peace.

Bell, T. D. (1999). Weaponization of space: Understanding strategic and technological inevitabilities. *Maxwell Air Force Base, Ala.* : Center for Strategy and Technology, Air War College, Air University, 1999.

Buono, S. (2020). Merely a "Scrap of Paper"? The Outer space treaty in historical perspective. *Diplomacy & Statecraft*, 31(2), 350–372. https://doi-org.ezproxy.umgc.edu/10.1080/09592296.2020.1760038

Burris, M. (2013). Astroimpolitic: organizing outer space by the sword. *Strategic Studies Quarterly*, *7*(3), 108–129. http://www.jstor.org/stable/26270588

CONSTANTIN-BERCEAN, I. (2022). US-China space race: From geopolitics to astropolitics. *Romanian Review of Political Sciences & International Relations*, *19*(2), 122–134.

Defense Intelligence Agency. (2022). *Challenges to security in space: Space reliance in an era of competition and expansion* (ISBN 978-0-16-095566-2). https://www.dia.mil/Portals/110/Documents/News/Military_Power_Publications/Challenges_Security_Space_2022.pdf

Grossman, K. (2000). Astro-Imperialism: War in space. *Earth Island Journal*, *15*(1), 40–41.

Gubrud, M. A. (2011). Chinese and US kinetic energy space weapons and arms control. *Asian Perspective*, *35*(4), 617–641.

Montluc, B. de. (2011). *A New International Strategic Context for Space Policies*. Nova Science Publishers, Inc.

NASA. (n.d.). *NASA: Artemis accords*. https://www.nasa.gov/specials/artemis-accords/index.html

Haines. (2022). *Chinese space activities will increasingly challenge US interests through 2030* (NIE 2021-01634). ODNI. https://www.dni.gov/files/ODNI/documents/assessments/NICM-Declassified-Chinese-Space-Activities-through-2030--2022.pdf

History.com Editors. (2009, October 14). *Arms race*. History.com. Retrieved March 16, 2023, from https://www.history.com/topics/cold-war/arms-race#arms-races-continue

Jackson, N.J. (2020). Russia's Space Security Policy. In: Schrogl, KU. (eds) Handbook of Space Security. Springer, Cham. https://doi.org/10.1007/978-3-030-23210-8_128

Pražák, J. (2021). Dual-use conundrum: Towards the weaponization of outer space? *Acta Astronautica*, *187*, 397–405. https://doi-org.ezproxy.umgc.edu/10.1016/j.actaastro.2020.12.051

Sönnichsen, A., & Lambach, D. (2020). A developing arms race in outer space? De-constructing the dynamics in the field of anti-satellite weapons. *Security & Peace / Sicherheit Und Frieden*, *38*(1), 5–9. https://doi-org.ezproxy.umgc.edu/10.5771/0175-274x-2020-1-5

The State Council Information Office of the People's Republic of China. (2022, January 28). *China's space program: A 2021 perspective*. 国家航天局.

Sheposh, R. (2023). Militarization of Space. *Salem Press Encyclopedia*.

United States Department of State. (2021, November 15). Russia conducts destructive anti-satellite missile test. https://www.state.gov/russia-conducts-destructive-anti-satellite-missile-test/

UNOOSA. (n.d.). Space law treaties and principles. *United Nations Office for Outer Space Affairs*. https://www.unoosa.org/oosa/en/ourwork/spacelaw/treaties.html#:~:text=The%20%22Outer%20Space%20Treaty%22,force%20on%2010%20October%201967

Whitford, G. (2019, October 29). *Trouble in the stars: The importance of US-China bilateral cooperation in space*. Harvard International Review. https://hir.harvard.edu/trouble-in-the-stars-the-importance-of-us-china-bilateral-cooperation-in-space/

XU, W., YAN, L., HU, Z., & LIANG, B. (2019). Area-oriented coordinated trajectory planning of dual-arm space robot for capturing a tumbling target. *Chinese Journal of Aeronautics*, *32*(9), 2151–2163. https://doi-org.ezproxy.umgc.edu/10.1016/j.cja.2019.03.018